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Mozzie: a computationally efficient simulator for the spatio-temporal modelling of mosquitoes

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Summary

`Mozzie` enables simulation of the lifecycle and spatial spread of mosquitoes. `Mozzie` can be used to assess risks associated with disease-control strategies at local, regional or continental scales. Most particularly, strategies involving genetic alterations of mosquitoes to eliminate malaria, are of prime interest.

More technically, `Mozzie` simulates a population-dynamics model that uses differential equations or delay differential equations (Bohner et al., 2018; El-Hachem & Beeton, 2024) to describe the spread and persistence of mosquitoes that may be genetic altered. Genetic alterations are flexibly modelled: these can involve any number of alleles; Mendelian or non-Mendelian inheritance, including gene drives; they can be self-limiting or self-sustaining; and can include the emergence of resistant alleles. The model allows simulation of N mosquito species. It incorporates mate-choice, hybridisation and intra-specific competition that occur within complexes of mosquito species (Beeton et al., 2020). This fills a gap that currently exists among similar models, allowing researchers to assess potential transfer of the genetic alterations between (sub-)species.

`Mozzie` supports spatial and temporal variations in lifecyle parameters, and local diffusion and wind-assisted, long range, advection. For example, wind patterns and the capacity of the landscape to support mosquitoes can vary spatially and temporally, reflecting daily variations, seasonality, and local conditions.

Conversely, `Mozzie` does not contain human agents, nor does it consider the effect of genetic control strategies on the prevalence of pathogens such as the malaria parasite, among human or animal populations.

`Mozzie` has been used by the authors to simulate the spread across sub-Saharan Africa of a theoretical, population-modifying, gene drive in *Anopheles gambiae s.s.* and *Anopheles coluzzii* (Beeton et al., 2022) (that paper also describes the mathematics of a particular mosquito lifecycle model that is contained in `Mozzie`). It has also been used to predict the spread of Target Malaria's Paternal Male Bias construct (Galizi et al., 2014) following a proposed field-release of genetically modified *Anopheles coluzzi* male mosquitoes in Burkina Faso (Hosack et al., 2023).

Statement of need

The numerical implementation of Mozzie is ecologically interpretable (Hosack et al., 2023) and computationally and I/O efficient. This allows rapid simulation at continental scales to investigate sensitivity to input parameters, as required in risk assessments. It is written in

Cython (Behnel et al., 2011) (a mixture of Python and C), and simulations are run using Python. The test coverage of the `Mozzie` codebase is over 99%, meaning it is also suitable for risk assessments that could be subject to considerable scrutiny.

Alternatives to `Mozzie` include:

- MIT's HYDREMATS software (Bomblies et al., 2009). HYDREMATS is a coupled hydrology and entomology model that uses an agent-based approach for mosquito-human dynamics, and focuses on high-resolution village-scale understanding of malaria without genetically-modified mosquitoes.
- The well-established SkeeterBuster focuses on the *Aedes aegypti* species in order to understand insecticidal control measures such as spraying (Gunning et al., 2022; Magori et al., 2009). A stochastic, mechanistic approach is employed.
- OpenMalaria (T. Smith et al., 2006) is an open-source C++ program enabling simulation of malaria epidemiology, typically at the village scale, in order to assess the efficacy of non-genetic malaria interventions.
- IDM's EMOD software can simulate malaria epidemiology using an agent-based approach, with spatial structure based on a network (Bershteyn et al., 2018). Less emphasis is spent on genetic modifications, and a single mosquito species is the focus.
- The dynamAedes R package can be used to study the spatio-temporal evolution of a single mosquito species, with particular attention paid to the impact of temperature heterogeneity.
- The exDE R package solves models of mosquito-borne pathogen dynamics and control (Wu et al., 2023). Attention is payed to sophisticated representations of mosquito lifecycles, including exogenous forcing by weather and vector control, as well as mosquito-malaria-human interatctions. Although a single mosquito species is the focus, the framework allows for multiple species. The code centers on traditional vector controls, rather than genetic controls.
- Berkeley's MGDrivE is an open-source framework to study gene-drives in mosquito populations (Mondal et al., 2024; Sánchez C. et al., 2019), which is written in R. With regards to lifecycle dynamics, MGDrivE has similar functionality to `Mozzie`, although MGDrivE focuses on single species, in contrast to `Mozzie` where transfer of genetic constructs between (sub-)species is of interest. MGDrivE's spatial structure is based on a network, where each node in the network could be thought of as a household, house block, or even a city. In terms of functionality, MGDrivE is the most similar to `Mozzie`, although the numerical methods employed are quite different.

If spatially explicit, these model spatial structure using a network. In contrast, `Mozzie` uses a continuous-space (diffusion-advection equation) approach, deliberately incorporating long-range dispersal in a way that is ecologically interpretable (Hosack et al., 2023). In addition, `Mozzie` does not focus on single species, but concentrates on the interaction of multiple (sub-)species. Many of the aforementioned alternatives contain human agents and the malaria parasite, which `Mozzie` does not.

In addition to the publicly available codes mentioned above, many academic articles consider the lifecycle and spatial spread of mosquitoes, for example (Bruzzone & Utgés, 2022; Dufourd & Dumont, 2013; Dye & Cain, 2024; Endo & Eltahir, 2018; Fang et al., 2020; Fernández-Carrión et al., 2018; Lutambi et al., 2013; North et al., 2013; Roques & Bonnefon, 2016; Silva et al., 2020; N. R. Smith et al., 2018; Yamashita, Takahashi, et al., 2018; Yamashita, Das, et al., 2018), but few have published their code. Most appear to rely on unpublished scripts in codes such as MATLAB (Fernández-Carrión et al., 2018; Lutambi et al., 2013; Yamashita, Das, et al., 2018), or concentrate on specialised scenarios (Bruzzone & Utgés, 2022; Roques & Bonnefon, 2016).

Example

Figure 1 shows results from `Mozzie` simulations when using 2 inter-breeding, hybridising and competing mosquito species (Beeton et al., 2022). A gene-drive is introduced into one of these species, and the modified individuals are released from one of 15 sites throughout sub-Saharan Africa.

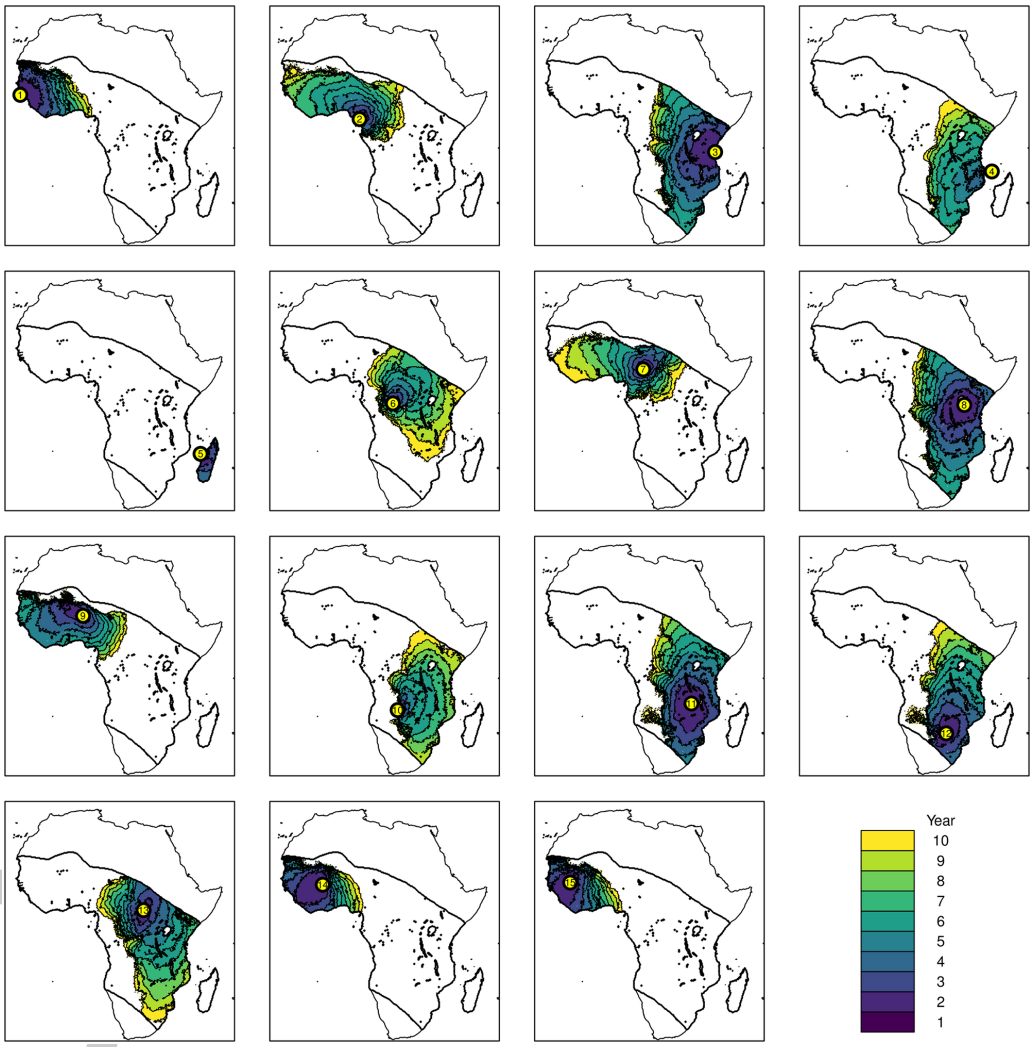

Figure 1: The invasion front of genetically-modified mosquito species, released from different points (Beeton et al., 2022). Figure used under the Creative Commons Attribution License.

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